

Terrain mapping flexible LiDAR sensor

The Leica TerrainMapper-2 linear-mode LiDAR airborne sensor for regional mapping projects delivers outstanding accuracy, unmatched point density, and



even-point distribution even in mountainous terrain thanks to gateless MPiA. It is designed to offer the maximum flexibility for all applications from narrow-swath corridors to high-altitude applications over complex and changing environments for the delivery of highest fidelity data. The TerrainMapper-2 includes a 2MHz LiDAR sensor combined with two nadir 150MP cameras in RGB and NIR. With the integration of the most innovative optical system on the market the sensor delivers high-quality images with every flight, even during long collection days and low sun angles. The sensor can easily be upgraded with four additional oblique cameras, turning the system into a 3D city-mapping machine.

Leica Geosystems
www.leica-geosystems.com

Low-cost yet efficient network accelerator

The MAX78000 low-power neural network accelerated microcontroller from Maxim executes artificial intelligence (AI) inferences at less than 1/100th the



energy of software solutions, dramatically improves run-time for battery-powered AI applications, while enabling complex new AI use cases previously considered impossible. With no compromise in latency or cost, the MAX78000 executes inferences at a fraction of

the cost of FPGA or GPU solutions. By integrating a dedicated neural network accelerator with a pair of microcontroller cores, the MAX78000 overcomes serious limitations, enabling machines to see and hear complex patterns with local, low-power AI processing that executes in real time. Applications such as machine vision, audio, and facial recognition can thus be made more efficient.

Maxim Integrated
www.maximintegrated.com

Real-time industrial 3D camera

Based on the latest generation of lasers and algorithms, this real-time industrial 3D camera from Orbbec provides exceptional 3D precision and accuracy. It can acquire a highly accurate image with full-colour texture at a working distance. The camera also offers improved picture



precision with a focus on real-time 3D vision processing and endurance in harsh environmental environments. The company engineers' expertise in machine learning, AI, depth algorithms, and smart computer vision helps to create compact standard solutions with technically mature functions for industrial applications.

Orbbec
www.orbbec3d.com

Deep learning chip for audio and sensor apps

Syntiant's NDP120 neural decision processor is a special-purpose chip for audio and sensor processing for always-on applications in battery-powered devices. It applies neural processing to run multiple applications simultaneously with minimal battery power consumption, including echo-cancellation, beamforming, noise suppression, speech enhancement, speaker identification, keyword spotting, multiple wake words, event detection, and local commands recognition. NDP120 brings always-on neural processing to all types of consumer prod-

ucts, including mobile phones, earbuds, wearables, smart speakers, laptops, smart home applications, and security devices. Ideal for use where audio filtering and echo cancellation are required for far-field speech processing, it packages a highly flexible, ultra-low-power deep



neural network inference engine, with a highly configurable audio front-end interface. It also supports multi-modal sensor fusion, including sensors for infrared detection, multi-axis acceleration, tilt, magnetic field, and pressure.

Syntiant
www.syntiant.com

Turnkey power module for energy harvesting

The Xidas EHM-UNIV-1 is a small, turnkey surface mount module designed to capture small amounts of harvested energy from such sources as photovoltaic cells, piezoelectric, electro-mechanical,



and thermoelectric generators, and continuously trickle charges storage elements like rechargeable Li-ion batteries, thin-film batteries, or conventional capacitors. It can be used to power small electronic devices and battery-free systems. When used with the Xidas's RHB-1530 small hybrid supercapacitor/lithium battery, it provides an ideal solution for driving wireless sensors. Its power management IC reduces the number of required external components and thus development cost. The signal conditioning provides a direct input for DC input sources (like photovoltaic cells) or built-in rectification circuitry for AC sources (like electromagnetic transducers). It can harvest energy from signals as low as 380mV.

Xidasiot
www.xidasiot.com